# LESSON PLAN

## NDT (19)

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| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 06 |
| **Total No. of Lesson Plans** | 9 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

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| **1. Topic** | Inspection of neutron radiography introduction, different methods principle and various applications |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Inspection of neutron radiography introduction, different methods principle and various applications |
| **5. Specific Objectives** | 19.1.3.1 Trainee will be able to explain the principle of radiographic NDT. |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

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| **7. Practical in Lab/Equipment Rooms/Simulators** | Review of theoretical concepts; Safety precautions regarding radiation shielding and source handling; Instructor supervision during Q&A; session |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

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| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What is the primary difference between X-ray and neutron radiography?
(b) Which materials are difficult to inspect using conventional X-ray methods?
(c) Define the term 'radiographic NDT'.

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Introduction to Neutron Radiography<br>• Basic Principle of Interaction<br>• Neutron Source Types<br>• Detection Systems Overview<br>• Image Formation Process | L & Ds | PPT, IB | 15 min |
| (b) | • Different Inspection Methods<br>• Transmission Technique Details<br>• Backscatter Technique Details<br>• Autoradiography Applications<br>• Digital Imaging Systems | L & Ds | PPT, PDF | 20 min |
| (c) | • Industrial Applications Overview<br>• Defense Sector Uses<br>• Material Characterization<br>• Flaw Detection Capabilities<br>• Limitations and Safety | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain how neutrons interact with light elements compared to heavy elements.
(b) Describe one specific application of neutron radiography in the defense sector.
(c) List two major limitations of neutron radiography compared to X-ray.

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| **12. Home Assignment and Follow Up** | Prepare a brief note comparing the penetration capabilities of neutrons versus X-rays for inspecting composite materials, referencing the next lesson on Ultrasonic Inspection. |

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| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | Ultra-sonic inspection, basic principle waves propagation characteristics acoustic impedance basic inspection methods, ultrasonic equipment, data representation, transducer, various probe couplant | Refer to the master syllabus reference list | LP 07 |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
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**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min